Subject Code 22ES1EE101	R22

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

B.Tech. I Year I Semester Regular/Supplementary Examinations, January 2025

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

(Common to CSE, IT, CyS and DS)

Time: 3 hours **Max. Marks: 60**

Answer ALL questions in PART-A.
Answer any ONE question from each unit in PART-B.

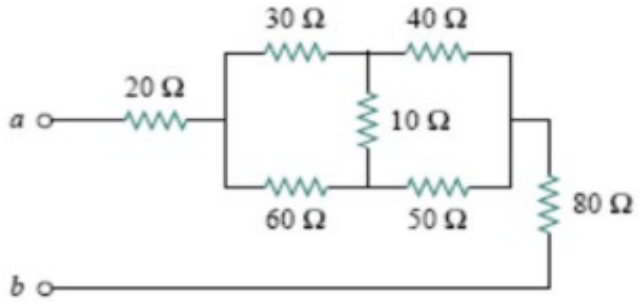
PART-A **5X2=10M**

a)	State Thevenin's theorem.	2M	CO-1	BL-1
b)	Define the following terms: (i) RMS value (ii) Average value	2M	CO-2	BL-1
c)	State the working principle of a transformer.	2M	CO-3	BL-2
d)	Differentiate the MCB and MCCB.	2M	CO-4	BL-2
e)	Define ripple factor.	2M	CO-5	BL-1

PART-B **5X10=50M**

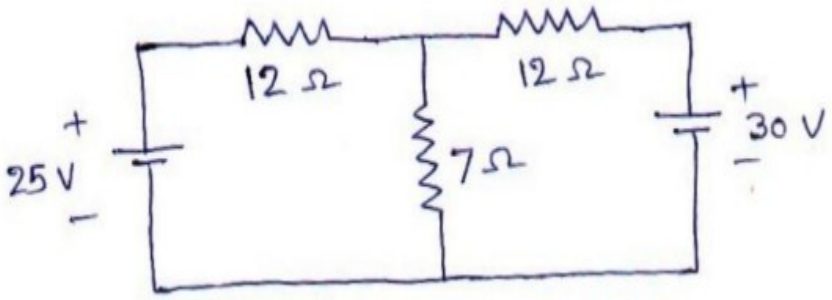
UNIT-I

1.	a)	State Kirchoff's laws with suitable examples.	5M	CO-1	BL-2
	b)	Calculate the equivalent resistance R_{ab} in the given circuit.	5M	CO-1	BL-3



OR

2.		Using Superposition theorem find the current through 7 Ω resistor as shown in figure.	10M	CO-1	BL-3



UNIT-II

3.	a)	Derive the line and phase voltage, current relationships in a three- phase star connected balanced load	5M	CO-2	BL-2
	b)	Derive form factor and Peak factor of a pure sine wave form.	5M	CO-2	BL-3

OR

4.	A series R-L-C circuit having a resistance of 7Ω, an inductance of 30mH and a capacitance of 100 μF is connected across a 200 V, 50 Hz supply. Calculate (i)Current (ii) Power factor (iii)Voltage across the Inductance and Capacitance and (iv)Active power v) Reactive power and vi) Apparent power.	10M	CO-2	BL-3

UNIT-III

5.	a) Derive the EMF equation of a single-phase transformer.	5M	CO-3	BL-2
	b) A 4-pole DC shunt motor is fed at 440V and takes an armature current of 50A. The resistance of armature circuit is 0.28Ω. The armature winding is wave connected with 888 conductors and useful flux per pole is 0.023wb. Calculate the speed of the motor.	5M	CO-3	BL-3

OR

6.	a) Derive the EMF equation of a DC Generator.	5M	CO-3	BL-2
	b) A 1-ϕ, 50 Hz, transformer has 120 turns on the primary winding and 200 turns on the secondary winding. The net cross-sectional area of the core is 100 cm2. If the primary winding is connected to 270V, 50 Hz supply, determine: i) the EMF induced in the secondary winding ii) The maximum value of the flux density in the core.	5M	CO-3	BL-3

UNIT-IV

7.	a) A 3-phase induction motor is wound for 4-pole and is supplied from 50Hz system. Calculate (i) the synchronous speed, (ii) the speed of the motor when slip is 4%.	5M	CO-4	BL-3
	b) Draw and explain the torque- slip characteristics of a 3 phase Induction motor.	5M	CO-4	BL-2

OR

8.	a) Classify different types of Batteries.	5M	CO-4	BL-3
	b) Explain about Pipe earthing.	5M	CO-4	BL-2

UNIT-V

9.	Describe working principle of PN diode with V-I characteristics.	10M	CO-5	BL-2

OR

10.	a) Illustrate the working of half Wave Rectifier with circuit diagram and draw it's waveforms.	5M	CO-5	BL-3
	b) Describe how the transistor acts as an amplifier.	5M	CO-5	BL-2

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